Analysis_tfm_spanish

March 1, 2026

1 EDA-PREPROCESO-MODELACIONES TFM

[9]:
```
#!pip install lightgbm
!pip install pandoc
```

```
Collecting pandoc
  Downloading pandoc-2.4.tar.gz (34 kB)
  Installing build dependencies … done
  Getting requirements to build wheel … done
  Preparing metadata (pyproject.toml) … done
Collecting plumbum (from pandoc)
  Downloading plumbum-1.10.0-py3-none-any.whl.metadata (8.4 kB)
Collecting ply (from pandoc)
  Downloading ply-3.11-py2.py3-none-any.whl.metadata (844 bytes)
Downloading plumbum-1.10.0-py3-none-any.whl (127 kB)
Downloading ply-3.11-py2.py3-none-any.whl (49 kB)
Building wheels for collected packages: pandoc
  Building wheel for pandoc (pyproject.toml) … done
  Created wheel for pandoc: filename=pandoc-2.4-py3-none-any.whl
size=34873
sha256=6cda3d5d64469680b05b729fe2f213bed14c6bbb65fa78ecc1c1f8dd7ba55aa8
  Stored in directory: /home/estebanm/.cache/pip/wheels/9c/2f/9f/b1aac8c3e74b4ee
327dc8c6eac5128996f9eadf586e2c0ba67
Successfully built pandoc
Installing collected packages: ply, plumbum, pandoc

3/3 [pandoc]       2/3 [pandoc]
Successfully installed pandoc-2.4 plumbum-1.10.0 ply-3.11
```

[2]:
```
# libraries
import pandas as pd
import numpy as np
import functools as ft
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.ensemble import IsolationForest
from sklearn.experimental import enable_iterative_imputer
from sklearn.impute import IterativeImputer
```

```
from sklearn.ensemble import RandomForestRegressor
# modelacion
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from sklearn.svm import LinearSVC, SVC
from sklearn.model_selection import cross_val_score
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import StackingClassifier
import lightgbm as lgb
# encoding
from sklearn.preprocessing import LabelEncoder, OneHotEncoder, OrdinalEncoder,␣
↪StandardScaler, MinMaxScaler

# seleccion de modelos
from sklearn.model_selection import GridSearchCV
from sklearn.model_selection import RandomizedSearchCV
from scipy.stats import uniform, randint

from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    confusion_matrix, classification_report, roc_curve, auc,
    roc_auc_score
)
from sklearn.feature_selection import (
    SelectKBest, f_classif, chi2, mutual_info_classif,
    RFE, SelectFromModel, VarianceThreshold
)

#otros
import time
import warnings
warnings.filterwarnings('ignore')
```

1.1 Carga de datos

[3]:
```
dt_librosa = pd.read_csv("/home/estebanm/Escritorio/docs_tfm/proyecto_tfm/
↪data_aumentada_castellano.csv")
#dt_librosa = pd.read_csv("/home/estebanm/Escritorio/docs_tfm/proyecto_tfm/
↪audio_features_ivanova_dataset.csv") # castellano

dt_librosa.head()
```

[3]:

```
   Unnamed: 0  dementia label gender   age  total_words  total_sentences  \
0           0         2    ad      M  57.0         71.0              4.0
1           1         2    ad      M  65.0         70.0              3.0
2           2         2    ad      M  67.0         77.0              4.0
3           3         2    ad      M  69.0         68.0              1.0
4           4         2    ad      M  72.0         66.0              3.0

   sentence_length_mean  sentence_length_std  type_token_ratio  …  \
0             17.750000            14.872374          0.690141  …
1             23.333333             3.399346          0.685714  …
2             19.250000             8.842370          0.714286  …
3             68.000000             0.000000          0.705882  …
4             22.000000            13.140269          0.742424  …

   pause_duration_mean  pause_time_ratio  speech_segment_duration_mean  \
0             0.329075          0.250182                      0.867556
1             0.344533          0.250000                      0.612211
2             0.384000          0.250217                      0.892690
3             0.337778          0.250219                      2.122000
4             0.468571          0.250203                      0.578000

   speech_segment_duration_cv  pause_duration_skewness  \
0                    0.703391                 2.069982
1                    0.645111                 1.558600
2                    0.718931                 3.689300
3                    1.420813                 0.457952
4                    0.515021                 1.403975

   pause_duration_kurtosis  voiced_ratio  mmse_df_info  schooling_years  \
0                 4.332522      0.708787          13.0             17.0
1                 1.956037      0.562693          28.0              NaN
2                15.348264      0.702867          23.0             17.0
3                -0.760515      0.931519          28.0             12.0
4                 1.385860      0.705452          19.0              8.0

   articulation_rate_wpm
0              64.482952
1             135.489676
2             167.357083
3             149.300752
4              67.159085

[5 rows x 51 columns]
```

[4]:

```
dt_librosa[['dementia', 'label']].value_counts()
```

[4]:
```
dementia  label
0         hc       1459
1         mci       523
2         ad        362
0         mci         5
1         hc          4
2         mci         3
0         ad          1
Name: count, dtype: int64
```

[5]:
```
dt_librosa.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2357 entries, 0 to 2356
Data columns (total 51 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Unnamed: 0                    2357 non-null   int64
 1   dementia                      2357 non-null   int64
 2   label                         2357 non-null   object
 3   gender                        2357 non-null   object
 4   age                           2357 non-null   float64
 5   total_words                   2357 non-null   float64
 6   total_sentences               2357 non-null   float64
 7   sentence_length_mean          2357 non-null   float64
 8   sentence_length_std           2357 non-null   float64
 9   type_token_ratio              2357 non-null   float64
 10  mattr_50                      2357 non-null   float64
 11  noun_ratio                    2357 non-null   float64
 12  content_verb_ratio            2357 non-null   float64
 13  adj_adv_ratio                 2357 non-null   float64
 14  function_word_ratio           2357 non-null   float64
 15  subordinate_sentence_ratio    2357 non-null   float64
 16  lexical_repetitions           2357 non-null   float64
 17  reformulations                2357 non-null   float64
 18  fillers                       2357 non-null   float64
 19  idea_density                  2357 non-null   float64
 20  local_coherence_bigram        2267 non-null   float64
 21  lexical_error_rate            2357 non-null   float64
 22  pronoun_to_propn_ratio        2357 non-null   float64
 23  mean_word_length              2357 non-null   float64
 24  readability                   2357 non-null   float64
 25  incomplete_sentence_ratio     2357 non-null   float64
 26  speech_rate_wpm               2357 non-null   float64
 27  f0_mean                       2357 non-null   float64
 28  f0_std                        2357 non-null   float64
 29  f0_range                      2357 non-null   float64
 30  loudness_mean                 2357 non-null   float64
```

```
 31  loudness_std                  2357 non-null   float64
 32  jitter_local                  2357 non-null   float64
 33  shimmer_local                 2357 non-null   float64
 34  hnr                           2357 non-null   float64
 35  alpha_ratio                   2357 non-null   float64
 36  hammarberg_index              2357 non-null   float64
 37  spectral_slope_mean           2357 non-null   float64
 38  spectral_slope_std            2357 non-null   float64
 39  duracion_s                    2357 non-null   float64
 40  pause_count                   2357 non-null   float64
 41  pause_duration_mean           2357 non-null   float64
 42  pause_time_ratio              2357 non-null   float64
 43  speech_segment_duration_mean  2357 non-null   float64
 44  speech_segment_duration_cv    2357 non-null   float64
 45  pause_duration_skewness       2357 non-null   float64
 46  pause_duration_kurtosis       2357 non-null   float64
 47  voiced_ratio                  2357 non-null   float64
 48  mmse_df_info                  2355 non-null   float64
 49  schooling_years               2342 non-null   float64
 50  articulation_rate_wpm         2357 non-null   float64
dtypes: float64(47), int64(2), object(2)
memory usage: 939.2+ KB
```

[6]:
```
dt_librosa.gender
```

[6]:
```
0       M
1       M
2       M
3       M
4       M
       ..
2352    M
2353    W
2354    W
2355    W
2356    M
Name: gender, Length: 2357, dtype: object
```

[7]:
```
dt_librosa.describe()
```

[7]:
```
         Unnamed: 0     dementia          age  total_words  total_sentences  \
count  2357.000000  2357.000000  2357.000000  2357.000000      2357.000000
mean   1178.000000     0.533305    76.860485    69.311105         3.277453
std     680.551615     0.747558     7.533849     6.817322         2.390828
min       0.000000     0.000000    50.000000    43.000000         0.914264
25%     589.000000     0.000000    71.000445    66.000000         1.788882
50%    1178.000000     0.000000    77.654854    67.083747         2.617194
75%    1767.000000     1.000000    82.343863    70.000000         4.555989
```

max	2356.000000	2.000000	96.000000	144.000000	30.000000

	sentence_length_mean	sentence_length_std	type_token_ratio	\
count	2357.000000	2357.000000	2357.000000	
mean	33.464285	5.150867	0.702931	
std	21.531436	4.813878	0.033201	
min	3.266667	-0.171225	0.450000	
25%	16.200000	0.498097	0.694219	
50%	28.349131	5.240948	0.711764	
75%	44.198288	7.348469	0.720304	
max	95.000000	27.500000	0.776119	

	mattr_50	noun_ratio	…	pause_duration_mean	pause_time_ratio	\
count	2357.000000	2357.000000	…	2357.000000	2357.000000	
mean	0.744476	0.300569	…	0.336303	0.250088	
std	0.031346	0.039270	…	0.089970	0.000199	
min	0.560000	0.137255	…	0.134261	0.249589	
25%	0.733171	0.272727	…	0.271444	0.249970	
50%	0.751236	0.304348	…	0.337607	0.250065	
75%	0.760250	0.330977	…	0.396885	0.250239	
max	0.819091	0.385714	…	0.657000	0.250597	

	speech_segment_duration_mean	speech_segment_duration_cv	\
count	2357.000000	2357.000000	
mean	0.849151	0.700635	
std	0.366809	0.129569	
min	0.325053	0.386939	
25%	0.613476	0.607869	
50%	0.751404	0.687857	
75%	0.957296	0.776520	
max	4.214857	1.424362	

	pause_duration_skewness	pause_duration_kurtosis	voiced_ratio	\
count	2357.000000	2357.000000	2357.000000	
mean	1.573770	3.175807	0.719023	
std	0.739086	3.886438	0.128968	
min	-0.378134	-1.465031	0.120301	
25%	1.105490	0.677622	0.667853	
50%	1.475772	2.016621	0.741172	
75%	1.945598	4.260383	0.805806	
max	4.894557	26.052424	0.967681	

	mmse_df_info	schooling_years	articulation_rate_wpm
count	2355.000000	2342.000000	2357.000000
mean	25.997863	9.090407	133.154173
std	4.089008	3.294647	38.579887
min	9.000000	-0.004906	31.208930

```
25%       24.000000         7.363995             106.449872
50%       27.478113         8.005982             134.271590
75%       28.971333        11.210603             161.310748
max       32.000000        19.000000             279.921686

[8 rows x 49 columns]
```

[8]:
```
dt_librosa.dementia.value_counts(normalize=True)
```

[8]:
```
dementia
0    0.621553
1    0.223589
2    0.154858
Name: proportion, dtype: float64
```

[9]:
```
dt_librosa['dementia'] = dt_librosa['dementia'].replace(2, 1)
```

[10]:
```
dt_librosa.dementia.value_counts(normalize=True)
```

[10]:
```
dementia
0    0.621553
1    0.378447
Name: proportion, dtype: float64
```

[11]:
```
# reemplazar valores negativos por cero
num_col = dt_librosa.select_dtypes(['int64', 'float64']).columns
dt_librosa[num_col] = dt_librosa[num_col].clip(lower=0)
```

1.2 Preprocesamiento

1.2.1 Valores perdidos-imputaciones

[12]:
```
dt_librosa.isnull().sum()
```

[12]:
```
Unnamed: 0                      0
dementia                        0
label                           0
gender                          0
age                             0
total_words                     0
total_sentences                 0
sentence_length_mean            0
sentence_length_std             0
type_token_ratio                0
mattr_50                        0
noun_ratio                      0
content_verb_ratio              0
adj_adv_ratio                   0
```

```
function_word_ratio              0
subordinate_sentence_ratio       0
lexical_repetitions              0
reformulations                   0
fillers                          0
idea_density                     0
local_coherence_bigram          90
lexical_error_rate               0
pronoun_to_propn_ratio           0
mean_word_length                 0
readability                      0
incomplete_sentence_ratio        0
speech_rate_wpm                  0
f0_mean                          0
f0_std                           0
f0_range                         0
loudness_mean                    0
loudness_std                     0
jitter_local                     0
shimmer_local                    0
hnr                              0
alpha_ratio                      0
hammarberg_index                 0
spectral_slope_mean              0
spectral_slope_std               0
duracion_s                       0
pause_count                      0
pause_duration_mean              0
pause_time_ratio                 0
speech_segment_duration_mean     0
speech_segment_duration_cv       0
pause_duration_skewness          0
pause_duration_kurtosis          0
voiced_ratio                     0
mmse_df_info                     2
schooling_years                 15
articulation_rate_wpm            0
dtype: int64
```

[13]:
```
dt_librosa.gender.value_counts()
```

[13]:
```
gender
W    1692
M     665
Name: count, dtype: int64
```

[14]:
```
dt_librosa.schooling_years.value_counts()
```

[14]:
```
schooling_years
8.000000     148
12.000000     63
6.000000      54
17.000000     32
3.000000      19
            …
14.855885      1
6.769518       1
8.002605       1
8.059822       1
16.237475      1
Name: count, Length: 2013, dtype: int64
```

[15]:
```python
dt_librosa.age.value_counts()
```

[15]:
```
age
82.000000    23
80.000000    21
81.000000    21
67.000000    20
76.000000    18
             ..
71.793668     1
83.841997     1
83.928842     1
68.338187     1
74.621324     1
Name: count, Length: 2043, dtype: int64
```

[16]:
```python
# porcentaje de valores perdidos
dt_librosa['gender'].isnull().sum()/len(dt_librosa) * 100
```

[16]:
```
0.0
```

[17]:
```python
# eliminar los valores perdidos
#dt_librosa = dt_librosa.dropna()
```

[18]:
```python
def imputation_nas(df, var):
    df[var] = df[var].fillna(df[var].median())
    return df[var]
```

[19]:
```python
imputation_nas(dt_librosa, 'local_coherence_bigram')
```

[19]:
```
0       0.000000
1       0.000000
2       0.000000
3       0.000067
```

```
4       0.000000
          …
2352    0.000000
2353    0.000000
2354    0.000100
2355    0.016532
2356    0.000057
Name: local_coherence_bigram, Length: 2357, dtype: float64
```

[20]:
```
imputation_nas(dt_librosa, 'schooling_years')
```

[20]:
```
0       17.000000
1        8.005982
2       17.000000
3       12.000000
4        8.000000
          …
2352    11.909458
2353     6.011416
2354    16.993945
2355     8.015323
2356    16.237475
Name: schooling_years, Length: 2357, dtype: float64
```

[21]:
```
imputation_nas(dt_librosa, 'mmse_df_info')
```

[21]:
```
0       13.000000
1       28.000000
2       23.000000
3       28.000000
4       19.000000
          …
2352    29.939767
2353    22.560554
2354    24.425479
2355    27.427152
2356    22.950260
Name: mmse_df_info, Length: 2357, dtype: float64
```

[22]:
```
# Identificar columnas categóricas
cat_cols = dt_librosa.select_dtypes(include=['object', 'category']).columns.
 ↪tolist()
# calcular la moda para cada columna categorica
modas = {}
for col in cat_cols:
    moda = dt_librosa[col].mode()
    if not moda.empty:
```

```
        modas[col] = moda[0]
        print(f"{col}: Moda = '{modas[col]}' ({dt_librosa[col].isna().sum()}␣
↪nulos a reemplazar)")
    else:
        modas[col] = None
        print(f"{col}: No hay moda, todos los valores son NAN's")

# imputar valores nulos
dt_librosa = dt_librosa.copy()
for col, moda_valor in modas.items():
    if moda_valor is not None:
        # contal nulos antes y despues
        nulos_antes = dt_librosa[col].isna().sum()
        dt_librosa[col] = dt_librosa[col].fillna(moda_valor)
        nulos_despues = dt_librosa[col].isna().sum()

        print(f"{col}: {nulos_antes}→ {nulos_despues} nulos ")
```

```
label: Moda = 'hc' (0 nulos a reemplazar)
gender: Moda = 'W' (0 nulos a reemplazar)
label: 0→ 0 nulos
gender: 0→ 0 nulos
```

[23]:
```
dt_librosa.isnull().sum()
```

[23]:
```
Unnamed: 0                     0
dementia                       0
label                          0
gender                         0
age                            0
total_words                    0
total_sentences                0
sentence_length_mean           0
sentence_length_std            0
type_token_ratio               0
mattr_50                       0
noun_ratio                     0
content_verb_ratio             0
adj_adv_ratio                  0
function_word_ratio            0
subordinate_sentence_ratio     0
lexical_repetitions            0
reformulations                 0
fillers                        0
idea_density                   0
local_coherence_bigram         0
lexical_error_rate             0
```

```
pronoun_to_propn_ratio          0
mean_word_length                0
readability                     0
incomplete_sentence_ratio       0
speech_rate_wpm                 0
f0_mean                         0
f0_std                          0
f0_range                        0
loudness_mean                   0
loudness_std                    0
jitter_local                    0
shimmer_local                   0
hnr                             0
alpha_ratio                     0
hammarberg_index                0
spectral_slope_mean             0
spectral_slope_std              0
duracion_s                      0
pause_count                     0
pause_duration_mean             0
pause_time_ratio                0
speech_segment_duration_mean    0
speech_segment_duration_cv      0
pause_duration_skewness         0
pause_duration_kurtosis         0
voiced_ratio                    0
mmse_df_info                    0
schooling_years                 0
articulation_rate_wpm           0
dtype: int64
```

[24]:
```
dt_librosa.columns
```

[24]:
```
Index(['Unnamed: 0', 'dementia', 'label', 'gender', 'age', 'total_words',
       'total_sentences', 'sentence_length_mean', 'sentence_length_std',
       'type_token_ratio', 'mattr_50', 'noun_ratio', 'content_verb_ratio',
       'adj_adv_ratio', 'function_word_ratio', 'subordinate_sentence_ratio',
       'lexical_repetitions', 'reformulations', 'fillers', 'idea_density',
       'local_coherence_bigram', 'lexical_error_rate',
       'pronoun_to_propn_ratio', 'mean_word_length', 'readability',
       'incomplete_sentence_ratio', 'speech_rate_wpm', 'f0_mean', 'f0_std',
       'f0_range', 'loudness_mean', 'loudness_std', 'jitter_local',
       'shimmer_local', 'hnr', 'alpha_ratio', 'hammarberg_index',
       'spectral_slope_mean', 'spectral_slope_std', 'duracion_s',
       'pause_count', 'pause_duration_mean', 'pause_time_ratio',
       'speech_segment_duration_mean', 'speech_segment_duration_cv',
       'pause_duration_skewness', 'pause_duration_kurtosis', 'voiced_ratio',
```

```
       'mmse_df_info', 'schooling_years', 'articulation_rate_wpm'],
      dtype='object')
```

[25]:
```
# eliminar algunas variables que no son utiles para el modelo
dt_lib = dt_librosa.drop(['Unnamed: 0', 'label'], axis=1) # 'sujeto',␣
↪'audio_file',
```

[26]:
```
# eliminar los valores perdidos
dt_lib= dt_lib.dropna()
```

[27]:
```
# dimensiones datos
dt_lib.shape
```

[27]:
```
(2357, 49)
```

[28]:
```
# revisar variable dependiente
dt_lib.dementia.value_counts(normalize=True)
```

[28]:
```
dementia
0    0.621553
1    0.378447
Name: proportion, dtype: float64
```

1.2.2 Transformaciones

[29]:
```
# genero
dt_lib.gender.value_counts()
```

[29]:
```
gender
W    1692
M     665
Name: count, dtype: int64
```

[30]:
```
# gender
encoded_df = pd.get_dummies(dt_lib[['gender']], prefix='gender',␣
↪drop_first=True)
dt_lib = pd.concat([dt_lib, encoded_df], axis=1)
```

[31]:
```
dt_lib = dt_lib.drop(['gender'], axis=1)
```

[32]:
```
# schooling_years
dt_lib['schooling_years'] = pd.cut(
    dt_lib['schooling_years'],
    bins=[0, 6, 8.5, 11, 19],
    labels=['basica', 'media', 'tecnica', 'universitaria'],
    include_lowest=True
)
```

[33]:
```
encoded_df = pd.get_dummies(dt_lib[['schooling_years']],␣
↪prefix='schooling_years', drop_first=True)
dt_lib = pd.concat([dt_lib, encoded_df], axis=1)
```

[34]:
```
dt_lib = dt_lib.drop(['schooling_years'], axis=1)
```

[35]:
```
# schooling_years
# crear orden
#capacidad_order = ['basica', 'media', 'tecnica', 'universitaria']
# crear encoder
#ordinal_encoder = OrdinalEncoder(categories=[capacidad_order])
# tranformar
#dt_lib['schooling_years'] = ordinal_encoder.
↪fit_transform(dt_lib[['schooling_years']])
```

[36]:
```
#dt_lib['schooling_years'].value_counts()
dt_lib.head()
```

[36]:
```
   dementia   age  total_words  total_sentences  sentence_length_mean  \
0         1  57.0         71.0              4.0             17.750000
1         1  65.0         70.0              3.0             23.333333
2         1  67.0         77.0              4.0             19.250000
3         1  69.0         68.0              1.0             68.000000
4         1  72.0         66.0              3.0             22.000000

   sentence_length_std  type_token_ratio  mattr_50  noun_ratio  \
0            14.872374          0.690141  0.714545    0.281690
1             3.399346          0.685714  0.714286    0.300000
2             8.842370          0.714286  0.755000    0.272727
3             0.000000          0.705882  0.727368    0.308824
4            13.140269          0.742424  0.794118    0.242424

   content_verb_ratio  …  speech_segment_duration_cv  \
0            0.126761  …                    0.703391
1            0.071429  …                    0.645111
2            0.064935  …                    0.718931
3            0.058824  …                    1.420813
4            0.060606  …                    0.515021

   pause_duration_skewness  pause_duration_kurtosis  voiced_ratio  \
0                 2.069982                 4.332522      0.708787
1                 1.558600                 1.956037      0.562693
2                 3.689300                15.348264      0.702867
3                 0.457952                 0.000000      0.931519
4                 1.403975                 1.385860      0.705452

   mmse_df_info  articulation_rate_wpm  gender_W  schooling_years_media  \
```

```
0          13.0              64.482952     False                   False
1          28.0             135.489676     False                    True
2          23.0             167.357083     False                   False
3          28.0             149.300752     False                   False
4          19.0              67.159085     False                    True

   schooling_years_tecnica  schooling_years_universitaria
0                    False                           True
1                    False                          False
2                    False                           True
3                    False                           True
4                    False                          False

[5 rows x 51 columns]
```

[37]:
```
# age
dt_lib['age'] = pd.cut(
    dt_lib['age'],
    bins=[50, 71, 77, 82, 96],
    labels=['adulto_q25', 'adulto_mediano', 'adulto_q75', 'adulto_max'],
    include_lowest=True
)
```

[38]:
```
encoded_df = pd.get_dummies(dt_lib[['age']], prefix='age', drop_first=True)
dt_lib = pd.concat([dt_lib, encoded_df], axis=1)
```

[39]:
```
dt_lib = dt_lib.drop(['age'], axis=1)
dt_lib.head()
```

[39]:
```
   dementia  total_words  total_sentences  sentence_length_mean  \
0         1         71.0              4.0             17.750000
1         1         70.0              3.0             23.333333
2         1         77.0              4.0             19.250000
3         1         68.0              1.0             68.000000
4         1         66.0              3.0             22.000000

   sentence_length_std  type_token_ratio  mattr_50  noun_ratio  \
0            14.872374          0.690141  0.714545    0.281690
1             3.399346          0.685714  0.714286    0.300000
2             8.842370          0.714286  0.755000    0.272727
3             0.000000          0.705882  0.727368    0.308824
4            13.140269          0.742424  0.794118    0.242424

   content_verb_ratio  adj_adv_ratio  …  voiced_ratio  mmse_df_info  \
0            0.126761       0.042254  …      0.708787          13.0
1            0.071429       0.085714  …      0.562693          28.0
2            0.064935       0.116883  …      0.702867          23.0
```

```
3            0.058824       0.117647  …      0.931519          28.0
4            0.060606       0.090909  …      0.705452          19.0

   articulation_rate_wpm  gender_W  schooling_years_media  \
0              64.482952     False                  False
1             135.489676     False                   True
2             167.357083     False                  False
3             149.300752     False                  False
4              67.159085     False                   True

   schooling_years_tecnica  schooling_years_universitaria  age_adulto_mediano  \
0                    False                           True               False
1                    False                          False               False
2                    False                           True               False
3                    False                           True               False
4                    False                          False                True

   age_adulto_q75  age_adulto_max
0           False           False
1           False           False
2           False           False
3           False           False
4           False           False

[5 rows x 53 columns]
```

[40]:
```
# age
# crear orden
#capacidad_order = ['adulto_q25', 'adulto_mediano', 'adulto_q75', 'adulto_max']
# crear encoder
#ordinal_encoder = OrdinalEncoder(categories=[capacidad_order])
# tranformar
#dt_lib['age'] = ordinal_encoder.fit_transform(dt_lib[['age']])
```

[41]:
```
#dt_lib['age'].value_counts()
```

[42]:
```
# eliminar los valores perdidos
dt_lib = dt_lib.dropna()
```

[43]:
```
dt_lib.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2357 entries, 0 to 2356
Data columns (total 53 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   dementia                      2357 non-null   int64
 1   total_words                   2357 non-null   float64
```

2	total_sentences	2357 non-null	float64
3	sentence_length_mean	2357 non-null	float64
4	sentence_length_std	2357 non-null	float64
5	type_token_ratio	2357 non-null	float64
6	mattr_50	2357 non-null	float64
7	noun_ratio	2357 non-null	float64
8	content_verb_ratio	2357 non-null	float64
9	adj_adv_ratio	2357 non-null	float64
10	function_word_ratio	2357 non-null	float64
11	subordinate_sentence_ratio	2357 non-null	float64
12	lexical_repetitions	2357 non-null	float64
13	reformulations	2357 non-null	float64
14	fillers	2357 non-null	float64
15	idea_density	2357 non-null	float64
16	local_coherence_bigram	2357 non-null	float64
17	lexical_error_rate	2357 non-null	float64
18	pronoun_to_propn_ratio	2357 non-null	float64
19	mean_word_length	2357 non-null	float64
20	readability	2357 non-null	float64
21	incomplete_sentence_ratio	2357 non-null	float64
22	speech_rate_wpm	2357 non-null	float64
23	f0_mean	2357 non-null	float64
24	f0_std	2357 non-null	float64
25	f0_range	2357 non-null	float64
26	loudness_mean	2357 non-null	float64
27	loudness_std	2357 non-null	float64
28	jitter_local	2357 non-null	float64
29	shimmer_local	2357 non-null	float64
30	hnr	2357 non-null	float64
31	alpha_ratio	2357 non-null	float64
32	hammarberg_index	2357 non-null	float64
33	spectral_slope_mean	2357 non-null	float64
34	spectral_slope_std	2357 non-null	float64
35	duracion_s	2357 non-null	float64
36	pause_count	2357 non-null	float64
37	pause_duration_mean	2357 non-null	float64
38	pause_time_ratio	2357 non-null	float64
39	speech_segment_duration_mean	2357 non-null	float64
40	speech_segment_duration_cv	2357 non-null	float64
41	pause_duration_skewness	2357 non-null	float64
42	pause_duration_kurtosis	2357 non-null	float64
43	voiced_ratio	2357 non-null	float64
44	mmse_df_info	2357 non-null	float64
45	articulation_rate_wpm	2357 non-null	float64
46	gender_W	2357 non-null	bool
47	schooling_years_media	2357 non-null	bool
48	schooling_years_tecnica	2357 non-null	bool
49	schooling_years_universitaria	2357 non-null	bool

```
 50  age_adulto_mediano              2357 non-null   bool
 51  age_adulto_q75                  2357 non-null   bool
 52  age_adulto_max                  2357 non-null   bool
dtypes: bool(7), float64(45), int64(1)
memory usage: 863.3 KB
```

1.2.3 Normalizacion

[44]:
```python
# Min-Max
scaler_minmax = MinMaxScaler()
num_cols = dt_lib.iloc[:, 1:].select_dtypes(include=['int64', 'float64']).
 ↪columns
dt_lib[num_cols] = scaler_minmax.fit_transform(dt_lib[num_cols])
dt_lib.describe()
```

[44]:
```
          dementia  total_words  total_sentences  sentence_length_mean  \
count  2357.000000  2357.000000      2357.000000           2357.000000
mean      0.378447     0.260506         0.081249              0.329189
std       0.485103     0.067498         0.082199              0.234718
min       0.000000     0.000000         0.000000              0.000000
25%       0.000000     0.227723         0.030070              0.140988
50%       0.000000     0.238453         0.058549              0.273428
75%       1.000000     0.267327         0.125207              0.446202
max       1.000000     1.000000         1.000000              1.000000

       sentence_length_std  type_token_ratio     mattr_50   noun_ratio  \
count          2357.000000       2357.000000  2357.000000  2357.000000
mean              0.187448          0.775577     0.712012     0.657307
std               0.174894          0.101806     0.120986     0.158053
min               0.000000          0.000000     0.000000     0.000000
25%               0.018113          0.748863     0.668380     0.545250
50%               0.190580          0.802662     0.738104     0.672516
75%               0.267217          0.828851     0.772894     0.779691
max               1.000000          1.000000     1.000000     1.000000

       content_verb_ratio  adj_adv_ratio  …  pause_count  \
count         2357.000000    2357.000000  …  2357.000000
mean             0.329373       0.438067  …     0.079789
std              0.106056       0.153735  …     0.076004
min              0.000000       0.000000  …     0.000000
25%              0.239581       0.367865  …     0.047473
50%              0.323404       0.423017  …     0.064932
75%              0.378026       0.519116  …     0.086637
max              1.000000       1.000000  …     1.000000

       pause_duration_mean  pause_time_ratio  speech_segment_duration_mean  \
count          2357.000000       2357.000000                   2357.000000
```

```
mean              0.386507          0.494911                      0.134737
std               0.172113          0.197097                      0.094300
min               0.000000          0.000000                      0.000000
25%               0.262431          0.378030                      0.074149
50%               0.389001          0.472470                      0.109607
75%               0.502401          0.645070                      0.162539
max               1.000000          1.000000                      1.000000

       speech_segment_duration_cv  pause_duration_skewness  \
count                 2357.000000              2357.000000
mean                    0.302380                 0.321627
std                     0.124895                 0.150791
min                     0.000000                 0.000000
25%                     0.212960                 0.225861
50%                     0.290062                 0.301513
75%                     0.375528                 0.397502
max                     1.000000                 1.000000

       pause_duration_kurtosis  voiced_ratio  mmse_df_info  \
count              2357.000000   2357.000000   2357.000000
mean                  0.124563      0.706557      0.739092
std                   0.146689      0.152196      0.177717
min                   0.000000      0.000000      0.000000
25%                   0.026010      0.646171      0.652174
50%                   0.077406      0.732695      0.803396
75%                   0.163531      0.808970      0.868278
max                   1.000000      1.000000      1.000000

       articulation_rate_wpm
count            2357.000000
mean                0.409891
std                 0.155118
min                 0.000000
25%                 0.302521
50%                 0.414384
75%                 0.523101
max                 1.000000

[8 rows x 46 columns]
```

1.3 Modelos

[45]:
```
#  Separar las variables X & y
X = dt_lib.drop('dementia', axis=1)
y = dt_lib['dementia']

print(f"\nX shape:{X.shape}")
```

```
print(f"\ny shape:{y.shape}")
print(f"n Distribucion de clases:")
print(y.value_counts(normalize=True))
```

```
X shape:(2357, 52)

y shape:(2357,)
n Distribucion de clases:
dementia
0    0.621553
1    0.378447
Name: proportion, dtype: float64
```

[46]:
```
# split dataset
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.3,␣
↪random_state=10, stratify=y)
print(f'Train set: {X_train.shape}')
print(f'Test set: {X_test.shape}')
print(f'\nChurn en train: {y_train.mean()*100:.1f}%')
print(f'Churn en test: {y_test.mean()*100:.1f}%')
```

```
Train set: (1649, 52)
Test set: (708, 52)

Churn en train: 37.8%
Churn en test: 37.9%
```

1.4 Seleccion de caracteristicas

[47]:
```
# SelectKBest con Anova F-test

k_best = 30
selector_kbest = SelectKBest(score_func=f_classif, k=k_best)
X_train_kbest = selector_kbest.fit_transform(X_train, y_train)
X_test_kbest = selector_kbest.transform(X_test)

print('FILTER METHOD: SelectKBest (ANOVA F-test)')
print('='*60)
print(f'K (features seleccionadas): {k_best}')
print(f'Features originales: {X_train.shape[1]}')
print(f'Features seleccionadas: {X_train_kbest.shape[1]}')

# obtener los scores
feature_scores = pd.DataFrame({
    'feature': X_train.columns,
    'score': selector_kbest.scores_
}).sort_values('score', ascending=False)
```

```python
print(f"\nTop {k_best} features por score F:")
print(feature_scores.head(k_best))

# Entrenar el modelo
model_kbest = RandomForestClassifier(n_estimators=100, random_state=42)
model_kbest.fit(X_train_kbest, y_train)
y_pred_kbest = model_kbest.predict(X_test_kbest)
accuracy_kbest = accuracy_score(y_test, y_pred_kbest)
print(f"\nAccuracy con selectKBest: {accuracy_kbest:.4f}")
```

```
FILTER METHOD: SelectKBest (ANOVA F-test)
============================================================
K (features seleccionadas): 30
Features originales: 52
Features seleccionadas: 30

Top 30 features por score F:
                          feature       score
43                   mmse_df_info  919.624529
21                speech_rate_wpm  232.059636
44          articulation_rate_wpm  230.749187
34                     duracion_s  193.860712
51                 age_adulto_max  101.518375
31               hammarberg_index   65.869189
36            pause_duration_mean   63.303365
25                  loudness_mean   61.241676
35                    pause_count   59.783867
23                         f0_std   59.432900
26                   loudness_std   55.163180
11            lexical_repetitions   52.385319
27                   jitter_local   48.911784
42                   voiced_ratio   42.979445
18               mean_word_length   42.580273
1                 total_sentences   37.210348
16             lexical_error_rate   32.966295
6                       noun_ratio   29.203433
7              content_verb_ratio   24.468760
8                   adj_adv_ratio   23.631850
20      incomplete_sentence_ratio   23.581642
0                     total_words   20.278915
13                        fillers   19.877611
48  schooling_years_universitaria   19.244236
46          schooling_years_media   18.975200
49             age_adulto_mediano   17.730900
5                        mattr_50   16.204834
24                       f0_range   12.441686
32            spectral_slope_mean   12.118068
```

```
12                  reformulations   10.142636

Accuracy con selectKBest: 0.9336
```

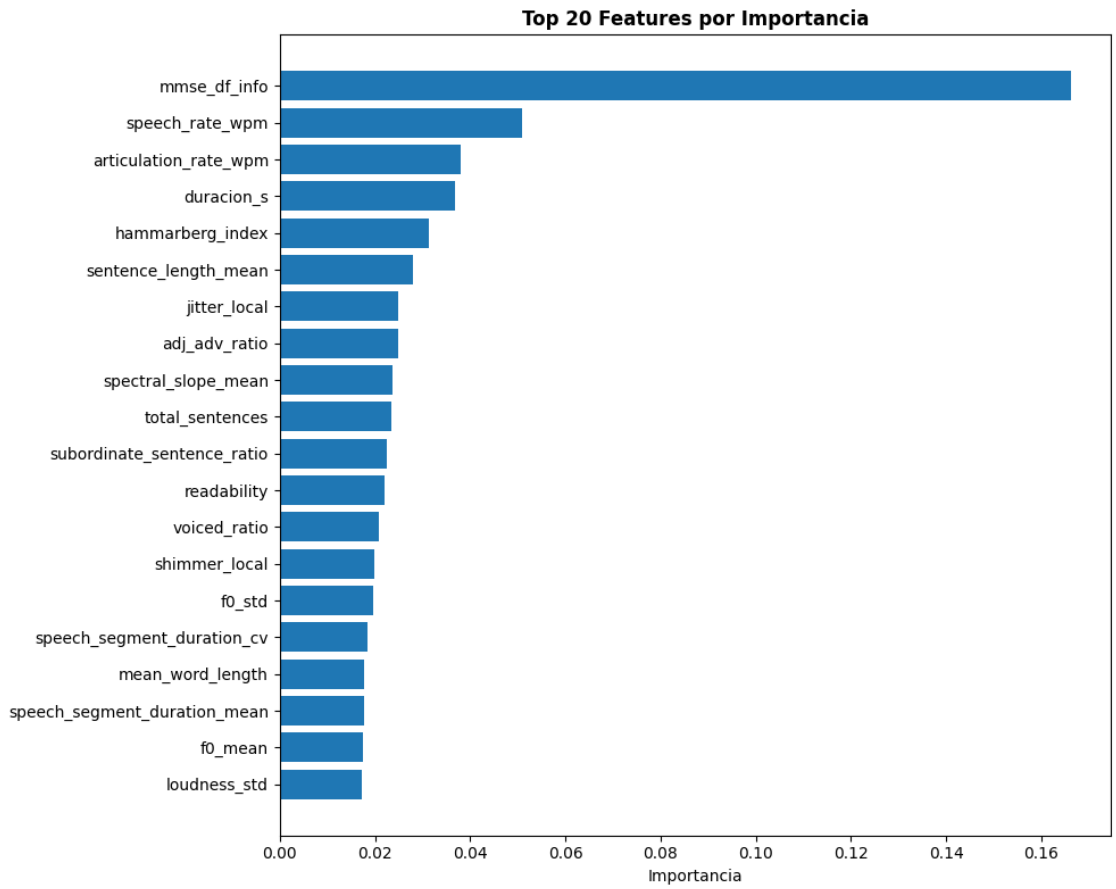
[48]:
```python
# Recursive Feature Elimination (RFE)

n_features_rfe = 25
estimator = LogisticRegression(max_iter=1000, random_state=23)
selector_rfe = RFE(estimator, n_features_to_select=n_features_rfe, step=1)
X_train_rfe = selector_rfe.fit_transform(X_train, y_train)
X_test_rfe = selector_rfe.transform(X_test)

print(f"Wrapper Method: Recursive Feature Elimination")
print(f"="*80)
print(f"Features seleccionadas: {n_features_rfe}")

# selecciondo features con su rankings
feature_ranking = pd.DataFrame({
    'feature': X_train.columns,
    'selected': selector_rfe.support_,
    'ranking': selector_rfe.ranking_
}).sort_values('ranking')

print("\nFeatures seleccionadas (ranking=1):")
print(list(feature_ranking[feature_ranking['ranking'] ==1]['feature']))

# entrenar al modelo
model_rfe = RandomForestClassifier(n_estimators=100, random_state=23)
model_rfe.fit(X_train_rfe, y_train)
y_pred_rfe = model_rfe.predict(X_test_rfe)
accuracy_rfe = accuracy_score(y_test, y_pred_rfe)
print(f"\nAccuracy con RFE:{accuracy_rfe:.4f}")
```

```
Wrapper Method: Recursive Feature Elimination
================================================================================
Features seleccionadas: 25

Features seleccionadas (ranking=1):
['total_words', 'schooling_years_universitaria', 'mmse_df_info', 'voiced_ratio',
'speech_segment_duration_cv', 'pause_duration_mean', 'spectral_slope_mean',
'hammarberg_index', 'alpha_ratio', 'hnr', 'shimmer_local', 'jitter_local',
'f0_std', 'speech_rate_wpm', 'incomplete_sentence_ratio',
'pronoun_to_propn_ratio', 'loudness_mean', 'local_coherence_bigram',
'total_sentences', 'sentence_length_mean', 'type_token_ratio',
'function_word_ratio', 'subordinate_sentence_ratio', 'lexical_repetitions',
'age_adulto_max']

Accuracy con RFE:0.9350
```

[49]:
```python
## Feature importance

rf_importance = RandomForestClassifier(n_estimators=100, random_state=23)
rf_importance.fit(X_train, y_train)

#obtener importancia
importances = pd.DataFrame({
    'feature': X_train.columns,
    'importance': rf_importance.feature_importances_
}).sort_values('importance', ascending=False)

print(f"\Embedded method: Feature importance")
print("="*80)
print(f"\nTop 20 features mas impornates:")
print(importances.head(20))

# visualizaciones
plt.figure(figsize=(10,8))
top_n = 20
plt.barh(range(top_n), importances['importance'].head(top_n))
plt.yticks(range(top_n), importances['feature'].head(top_n))
plt.xlabel('Importancia')
plt.title(f"Top {top_n} Features por Importancia", fontweight='bold')
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```

```
\Embedded method: Feature importance
================================================================================

Top 20 features mas impornates:
                         feature  importance
43                  mmse_df_info    0.166299
21                speech_rate_wpm    0.050951
44          articulation_rate_wpm    0.038013
34                     duracion_s    0.036823
31               hammarberg_index    0.031288
2            sentence_length_mean    0.027984
27                   jitter_local    0.024988
8                   adj_adv_ratio    0.024755
32            spectral_slope_mean    0.023736
1                total_sentences    0.023560
10    subordinate_sentence_ratio    0.022437
19                    readability    0.022075
42                   voiced_ratio    0.020895
28                  shimmer_local    0.019804
23                         f0_std    0.019680
39    speech_segment_duration_cv    0.018318
```

```
18              mean_word_length    0.017808
38  speech_segment_duration_mean    0.017687
22                       f0_mean    0.017528
26                  loudness_std    0.017312
```



1.5 Seleccion de caracteristicas para los modelos

[50]:
```
dt_lib = dt_lib[['total_words', 'mmse_df_info', 'voiced_ratio',␣
↪'speech_segment_duration_cv',
                 'pause_duration_mean', 'spectral_slope_mean',␣
↪'hammarberg_index', 'hnr', 'shimmer_local',
                 'f0_std', 'incomplete_sentence_ratio',␣
↪'pronoun_to_propn_ratio', 'loudness_mean', 'age_adulto_max',
                 'sentence_length_mean', 'total_sentences',␣
↪'function_word_ratio', 'subordinate_sentence_ratio',
                 'type_token_ratio', 'lexical_repetitions',
                 'schooling_years_universitaria', 'schooling_years_media',␣
↪'age_adulto_mediano', 'dementia']]
```

```
#'schooling_years_universitaria', 'schooling_years_media', 'age_adulto_mediano'


# HOT ENCODER
#dt_lib = dt_lib[['age', 'loudness_mean', 'pronoun_to_propn_ratio',␣
↪'shimmer_local', 'local_coherence_bigram', 'hnr',
#                  'hammarberg_index', 'lexical_repetitions',␣
↪'incomplete_sentence_ratio', 'function_word_ratio',
#                  'subordinate_sentence_ratio', 'pause_duration_mean',␣
↪'speech_segment_duration_cv', 'voiced_ratio',
#                  'type_token_ratio', 'mmse_df_info', 'total_sentences',␣
↪'total_words', 'spectral_slope_mean', 'f0_std',
#                 'schooling_years', 'dementia']]
```

[51]:
```
dt_lib.shape
```

[51]: (2357, 24)

[52]:
```
#  Separar las variables X & y
X = dt_lib.drop('dementia', axis=1)
y = dt_lib['dementia']

print(f"\nX shape:{X.shape}")
print(f"\ny shape:{y.shape}")
print(f"n Distribucion de clases:")
print(y.value_counts(normalize=True))
```

```
X shape:(2357, 23)

y shape:(2357,)
n Distribucion de clases:
dementia
0    0.621553
1    0.378447
Name: proportion, dtype: float64
```

[53]:
```
# split dataset
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.3,␣
↪random_state=10, stratify=y)
print(f'Train set: {X_train.shape}')
print(f'Test set: {X_test.shape}')
print(f'\nChurn en train: {y_train.mean()*100:.1f}%')
print(f'Churn en test: {y_test.mean()*100:.1f}%')
```

```
Train set: (1649, 23)
Test set: (708, 23)
```

```
Churn en train: 37.8%
Churn en test: 37.9%
```

[ ]:

```

```

1.5.1 Balanceo de datos

[54]:

```
#from imblearn.over_sampling import SMOTE

# Inicializar SMOTE con parámetros adecuados para un dataset pequeño
#smote = SMOTE(
#    random_state=42,
#    k_neighbors=3,  # Reducir vecinos porque tenemos pocos datos
#    sampling_strategy=0.80  # Aumentar la clase minoritaria al 60% de la␣
 ↪mayoritaria (no necesariamente 50-50)
#)

# Aplicar SMOTE solo a los datos de entrenamiento
#X_resampled, y_resampled = smote.fit_resample(X_train, y_train)

# Verificar la nueva distribución
#print("Distribución después de SMOTE:")
#print(pd.Series(y_resampled).value_counts())
#print(pd.Series(y_resampled).value_counts(normalize=True))
```

1.5.2 Regresión Logística

[55]:

```
logreg = LogisticRegression(random_state=23)

rejilla_parametros = {'C': np.linspace(0.1, 10, 100), 'penalty':['l2', 'l1']}
grid_lr = GridSearchCV(logreg, rejilla_parametros)
model_lr = grid_lr.fit(X_train, y_train)
model_lr.best_params_
```

[55]: {'C': 1.1, 'penalty': 'l2'}

[56]:

```
grid_logreg_pred = model_lr.best_estimator_.predict(X_test)

accuracy_logreg = accuracy_score(y_test, grid_logreg_pred)
y_prods_logreg = model_lr.predict_proba(X_test)[:,1]
```

[57]:

```
print(f"Accuracy {accuracy_logreg}")
```

```
Accuracy 0.8192090395480226
```

[58]:

```
# Matriz confusion
cm_logreg = confusion_matrix(y_test, grid_logreg_pred)
```

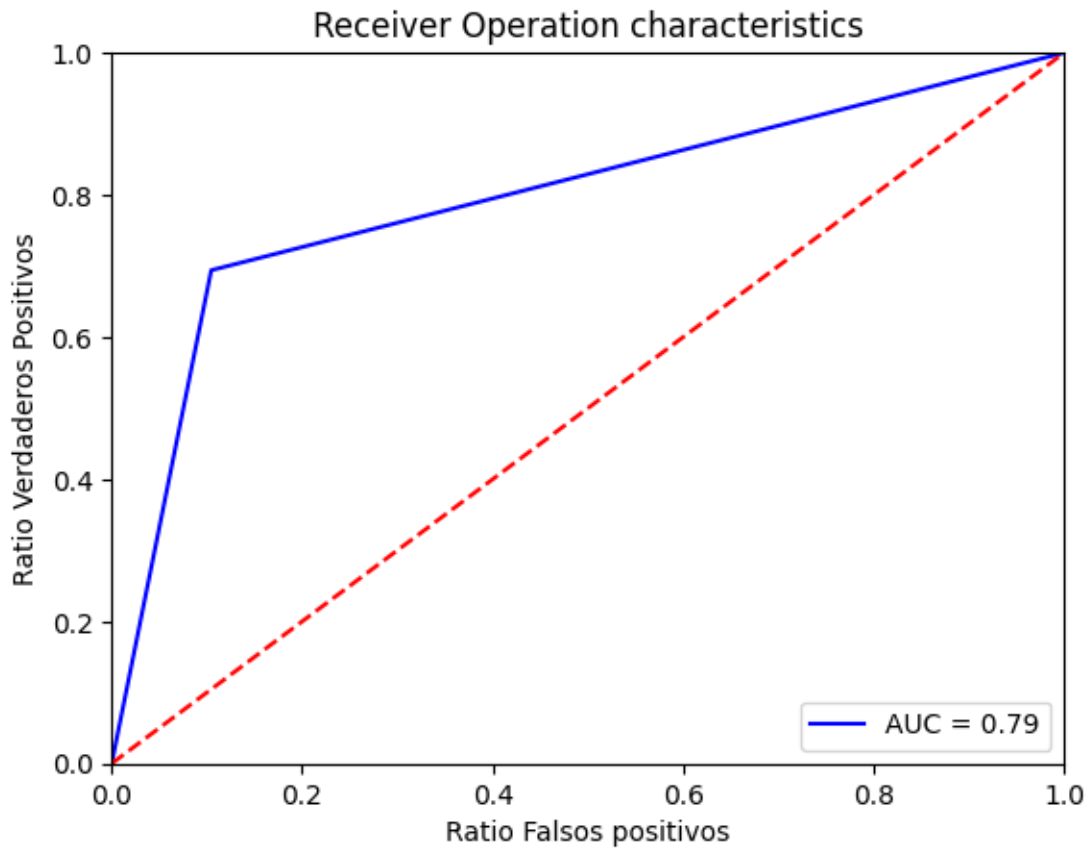
```
print(f"Matrix de confusion:")
print(cm_logreg)
```

```
Matrix de confusion:
[[394  46]
 [ 82 186]]
```

[59]:
```
# Reporte de classificacion
print(f"\nReporte de clasificacion:")
print(classification_report(y_test, grid_logreg_pred))
```

```
Reporte de clasificacion:
              precision    recall  f1-score   support

           0       0.83      0.90      0.86       440
           1       0.80      0.69      0.74       268

    accuracy                           0.82       708
   macro avg       0.81      0.79      0.80       708
weighted avg       0.82      0.82      0.82       708
```

[60]:
```
## Area bajo la curva (AUC)
fpr, tpr, threshold = roc_curve(y_test, grid_logreg_pred)
roc_auc = roc_auc_score(y_test, grid_logreg_pred)

# plot

plt.title("Receiver Operation characteristics")
plt.plot(fpr, tpr, 'b', label="AUC = %0.2f" % roc_auc)
plt.legend(loc = "lower right")
plt.plot([0, 1], [0, 1], 'r--')
plt.xlim([0,1])
plt.ylim([0,1])
plt.ylabel("Ratio Verdaderos Positivos")
plt.xlabel("Ratio Falsos positivos")
plt.show()
```



1.5.3 Arboles decisiones

[61]:
```
param_grid = {
    'max_depth': [2, 3, 10, 15, 20],
    'min_samples_split': [2, 5, 10, 15, 20],
    'criterion': ['gini', 'entropy']
}

dectree = GridSearchCV(estimator=DecisionTreeClassifier(),
                       param_grid = param_grid, cv=5,
                       scoring='accuracy')

model_tree = dectree.fit(X_train, y_train)

print(f"Best hyperparmeters: {model_tree.best_params_}")
print(f"Best score: {model_tree.best_score_}")
```

```
Best hyperparmeters: {'criterion': 'gini', 'max_depth': 10, 'min_samples_split':
2}
Best score: 0.870833563599521
```

[62]:
```
y_pred_dectree= model_tree.best_estimator_.predict(X_test)
accuracy_dectree = accuracy_score(y_test, y_pred_dectree)
y_prods_dectree = model_tree.predict_proba(X_test)[:,1]
```

[63]:
```
print(f"Accuracy {accuracy_dectree}")
```

```
Accuracy 0.8545197740112994
```

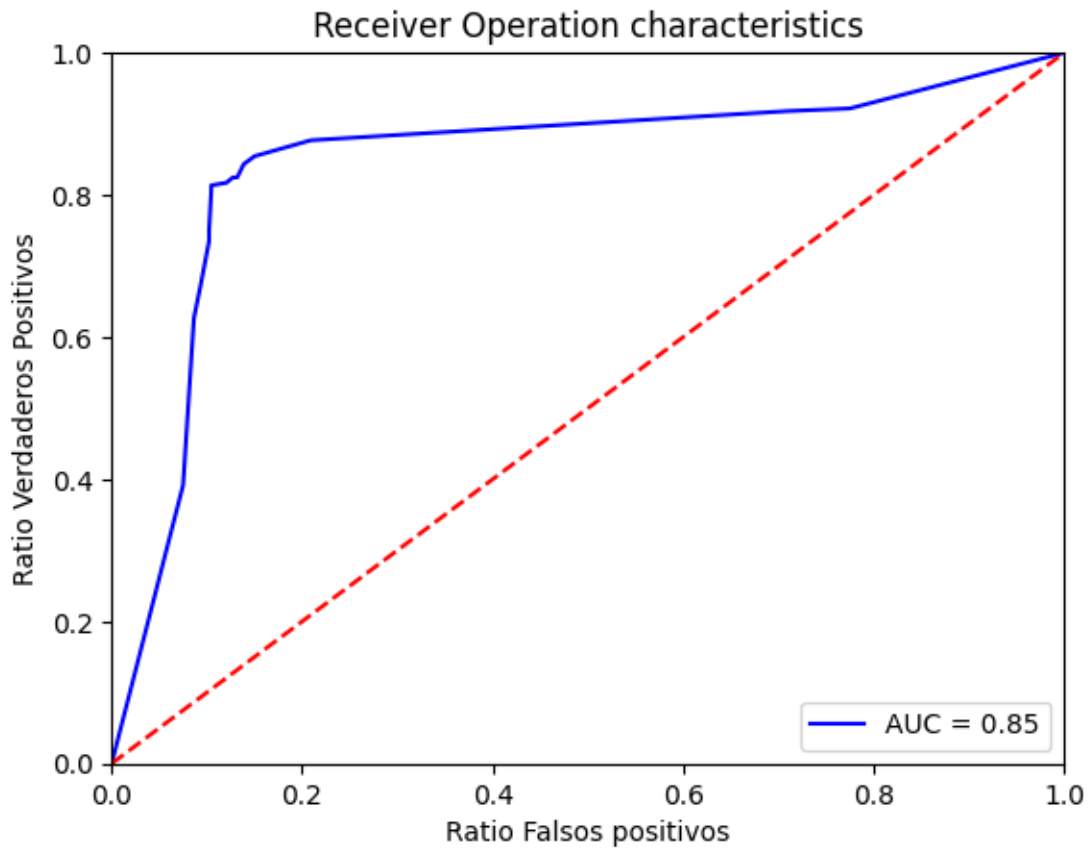
[64]:
```
# Matriz confusion
cm_dectree= confusion_matrix(y_test, y_pred_dectree)
print(f"\nMatrix de confusion:")
print(cm_dectree)
```

```
Matrix de confusion:
[[384  56]
 [ 47 221]]
```

[65]:
```
# Reporte de classificacion
print(f"\nReporte de clasificacion:")
print(classification_report(y_test, y_pred_dectree))
```

```
Reporte de clasificacion:
              precision    recall  f1-score   support

           0       0.89      0.87      0.88       440
           1       0.80      0.82      0.81       268

    accuracy                           0.85       708
   macro avg       0.84      0.85      0.85       708
weighted avg       0.86      0.85      0.85       708
```

[66]:
```
## Area bajo la curva (AUC)
fpr, tpr, threshold = roc_curve(y_test, y_prods_dectree)
roc_auc = roc_auc_score(y_test, y_prods_dectree)

# plot

plt.title("Receiver Operation characteristics")
plt.plot(fpr, tpr, 'b', label="AUC = %0.2f" % roc_auc)
plt.legend(loc = "lower right")
plt.plot([0, 1], [0, 1], 'r--')
plt.xlim([0,1])
plt.ylim([0,1])
plt.ylabel("Ratio Verdaderos Positivos")
plt.xlabel("Ratio Falsos positivos")
plt.show()
```



1.5.4 Random Forest

rf = RandomForestClassifier() gr_space = { 'max_depth': [5, 10, 15, 20, 25], # Añadir None y más opciones 'n_estimators': [50, 100, 200, 300], # Valores más altos 'max_features': ['sqrt', 'log2', 0.3, 0.5, 0.7], # Métodos automáticos y proporciones 'min_samples_leaf': [1, 2, 8, 10], 'min_samples_split': [2, 5, 10, 15], 'criterion': ['gini', 'entropy'], 'bootstrap': [True, False], 'max_samples': [0.7, 0.9, 1] # Para submuestreo } grid_rf = GridSearchCV(rf, gr_space, cv=3, scoring='roc_auc') model_grid_rf = grid_rf.fit(X_train, y_train)

print(f'Best hyperparameters are {model_grid_rf.best_params_}') print(f'Best score is: {model_grid_rf.best_score_}')

[69]:
```
rf = RandomForestClassifier()
param_dist = {
    'n_estimators': randint(50, 500),
    'max_depth': [5, 10, 15, 20, 25, None],
    'max_features': uniform(0.1, 0.9),  # Proporción
    'min_samples_split': randint(2, 20),
    'min_samples_leaf': randint(1, 10),
}
```

```
grid_rf= RandomizedSearchCV(
    rf, param_dist, n_iter=100, cv=5, scoring='roc_auc', n_jobs=-1,␣
 ↪random_state=42
)
model_grid_rf = grid_rf.fit(X_train, y_train)

print(f'Best hyperparameters are {model_grid_rf.best_params_}')
print(f'Best score is: {model_grid_rf.best_score_}')
```

```
Best hyperparameters are {'max_depth': None, 'max_features':
0.12974565961049356, 'min_samples_leaf': 1, 'min_samples_split': 2,
'n_estimators': 452}
Best score is: 0.9746557356412275
```

[70]:
```
y_pred_rf = model_grid_rf.predict(X_test)
accuracy_rf = accuracy_score(y_test, y_pred_rf)
y_prods_rf = grid_rf.predict_proba(X_test)[:,1]
```

[71]:
```
print(f"Accuracy {accuracy_rf}")
```

```
Accuracy 0.9336158192090396
```

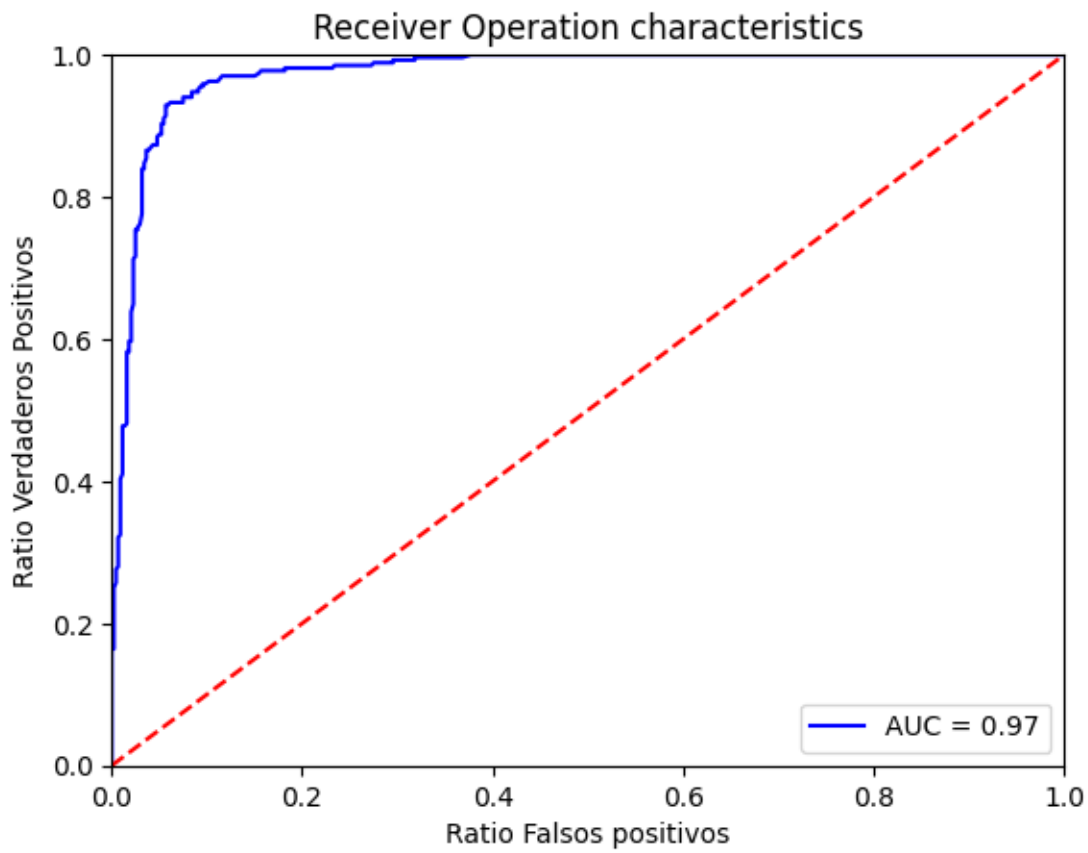
[72]:
```
# Matriz confusion
cm_rf = confusion_matrix(y_test, y_pred_rf)
print(f"\nMatrix de confusion:")
print(cm_rf)
```

```
Matrix de confusion:
[[415  25]
 [ 22 246]]
```

[73]:
```
# Reporte de classificacion
print(f"\nReporte de clasificacion:")
print(classification_report(y_test, y_pred_rf))
```

```
Reporte de clasificacion:
              precision    recall  f1-score   support

           0       0.95      0.94      0.95       440
           1       0.91      0.92      0.91       268

    accuracy                           0.93       708
   macro avg       0.93      0.93      0.93       708
weighted avg       0.93      0.93      0.93       708
```

[74]:
```
## Area bajo la curva (AUC)
fpr, tpr, threshold = roc_curve(y_test, y_prods_rf)
roc_auc = roc_auc_score(y_test, y_prods_rf)

# plot

plt.title("Receiver Operation characteristics")
plt.plot(fpr, tpr, 'b', label="AUC = %0.2f" % roc_auc)
plt.legend(loc = "lower right")
plt.plot([0, 1], [0, 1], 'r--')
plt.xlim([0,1])
plt.ylim([0,1])
plt.ylabel("Ratio Verdaderos Positivos")
plt.xlabel("Ratio Falsos positivos")
plt.show()
```



1.5.5 XGBoost

[75]:
```
xgb = XGBClassifier()
param_grid = {
    'max_depth': [3, 5, 7, 9, 10, 12],
    'learning_rate': [0.1, 0.01, 0.001],
    'subsample': [0.5, 0.7, 1]
}
grid_xgb = GridSearchCV(xgb, param_grid, cv=3, scoring='accuracy')
model_grid_xgb = grid_xgb.fit(X_train, y_train)

print(f'Best hyperparameters are {model_grid_xgb.best_params_}')
print(f'Best score is: {model_grid_xgb.best_score_}')
```

```
Best hyperparameters are {'learning_rate': 0.1, 'max_depth': 7, 'subsample':
0.7}
Best score is: 0.9114555389965225
```

[76]:
```
y_pred_xgb = model_grid_xgb.predict(X_test)
accuracy_xgb = accuracy_score(y_test, y_pred_xgb)
y_prods_xgb = model_grid_xgb.predict_proba(X_test)[:,1]
```

[77]:
```
print(f"Accuracy {accuracy_xgb}")
```

```
Accuracy 0.9336158192090396
```

[78]:
```
# Matriz confusion
cm_xgb = confusion_matrix(y_test, y_pred_xgb)
print(f"\nMatrix de confusion:")
print(cm_xgb)
```

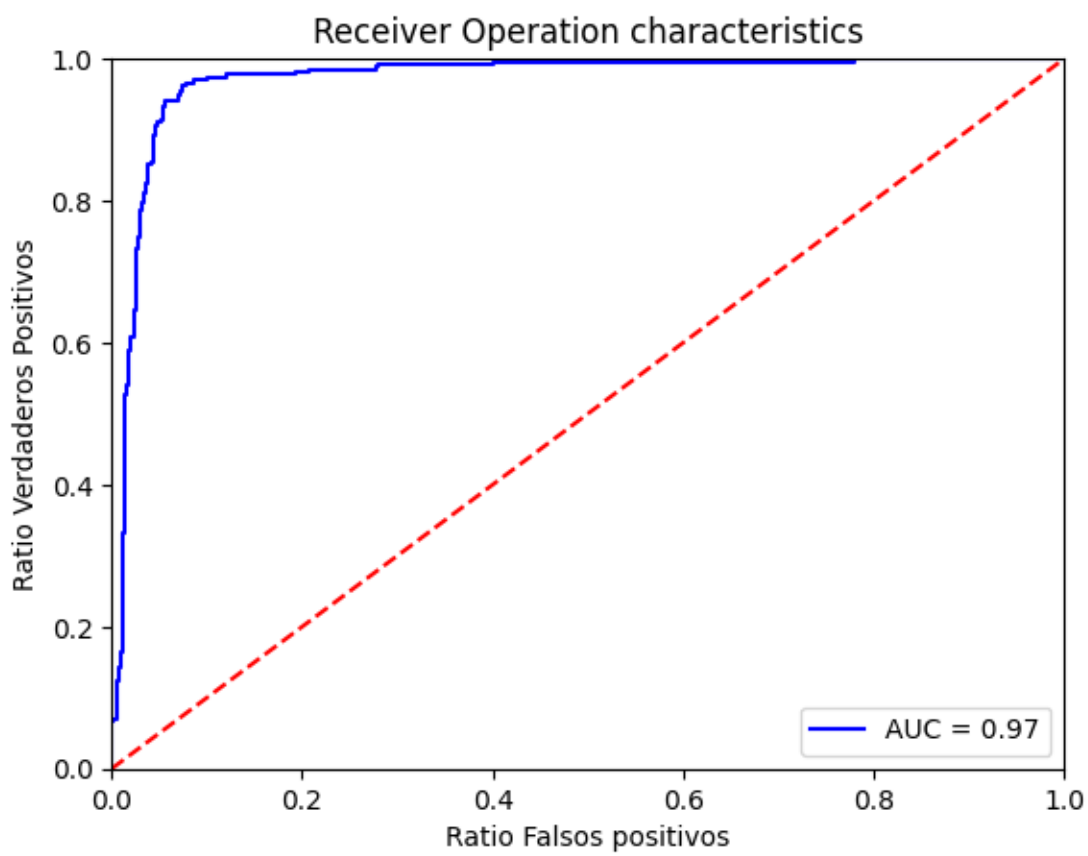
```
Matrix de confusion:
[[416  24]
 [ 23 245]]
```

[79]:
```
# Reporte de classificacion
print(f"\nReporte de clasificacion:")
print(classification_report(y_test, y_pred_xgb))
```

```
Reporte de clasificacion:
              precision    recall  f1-score   support

           0       0.95      0.95      0.95       440
           1       0.91      0.91      0.91       268

    accuracy                           0.93       708
   macro avg       0.93      0.93      0.93       708
```

```
weighted avg       0.93      0.93      0.93       708
```

[80]:
```python
## Area bajo la curva (AUC)
fpr, tpr, threshold = roc_curve(y_test, y_prods_xgb)
roc_auc = roc_auc_score(y_test, y_prods_xgb)

# plot

plt.title("Receiver Operation characteristics")
plt.plot(fpr, tpr, 'b', label="AUC = %0.2f" % roc_auc)
plt.legend(loc = "lower right")
plt.plot([0, 1], [0, 1], 'r--')
plt.xlim([0,1])
plt.ylim([0,1])
plt.ylabel("Ratio Verdaderos Positivos")
plt.xlabel("Ratio Falsos positivos")
plt.show()
```



1.5.6 LGBM

[81]:
```
from sklearn.model_selection import RandomizedSearchCV, train_test_split
from scipy.stats import uniform, randint, loguniform
from sklearn.metrics import average_precision_score

pos_rate = y_train.mean()
scale_pos_weight = (1-pos_rate)/ pos_rate

lgbm = lgb.LGBMClassifier(objective="binary",
                          metric="average_precision",
                          random_state=42,
                          n_jobs=-1,
                          is_unbalance=False,
                          scale_pos_weight= scale_pos_weight,
                         #early_stopping_round=60
                         )




param_dist= {
    "num_leaves": randint(8,64),
    "max_depth": randint(3, 9),
    "learning_rate": loguniform(0.003, 0.12),
    "n_estimators": randint(100, 1000),
    "subsample": uniform(0.6, 0.39),
    "colsample_bytree": uniform(0.5, 0.49),
    "min_child_samples": randint(5,30),
    "min_child_wight": uniform(0.001, 5),
    "reg_alpha": uniform(0.0, 8.0),
    "reg_lambda": uniform(0.5, 15.0),
}

#grid_lgbm = GridSearchCV(xgb, param_grid, cv=3, scoring='roc_auc')
search_lgbm = RandomizedSearchCV(lgbm,
                                 param_dist,
                                 n_iter=80,
                                 cv=5,
                                 scoring='average_precision',
                                 n_jobs=-1,
                                 random_state=42,
                                 return_train_score=True,
                                 error_score='raise'
                                )
model_search_lgbm = search_lgbm.fit(X_train, y_train)

print(f'Best hyperparameters are {model_search_lgbm.best_params_}')
```

```python
print(f'Best score is: {model_search_lgbm.best_score_}')
```

```
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.008706 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001527 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003379 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001175 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002768 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003490 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004108 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.005371 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.006121 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002961 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002306 seconds.
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You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002773 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002448 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003672 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003940 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002006 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
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[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
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You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
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-inf[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002434 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001241 seconds.
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You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.001620 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001871 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003760 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.005360 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001860 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002654 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.005138 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001199 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004513 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003152 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003986 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
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testing was 0.002628 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.013479 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002672 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002740 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.005353 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.005828 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001475 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002782 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002009 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001349 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001913 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
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[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004268 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002481 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002125 seconds.
You can set `force_col_wise=true` to remove the overhead.
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[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
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[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003334 seconds.
You can set `force_col_wise=true` to remove the overhead.
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[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001214 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001243 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001088 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001656 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.006121 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001404 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003439 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001112 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003738 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003475 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004057 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.007304 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003608 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004524 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001926 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
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[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002406 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001287 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004109 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001195 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002816 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003319 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
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testing was 0.002073 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001861 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001188 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003192 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001149 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004991 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002395 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001625 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001246 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001565 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004930 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002402 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.006330 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002065 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001670 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002658 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002837 seconds.
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You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003504 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Start training from score -0.494696
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001483 seconds.
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You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003609 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
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testing was 0.004554 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001992 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001859 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002013 seconds.
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You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.007577 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003321 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001135 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001171 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001522 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001621 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001754 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003388 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.000969 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001360 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
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[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001080 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001557 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002157 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001674 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.005698 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001505 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004577 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002006 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004049 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001677 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001037 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
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[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002634 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001726 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002843 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001543 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
```

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[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001585 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
```

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[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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the split requirements
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002003 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001370 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
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[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001360 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002823 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003145 seconds.
You can set `force_col_wise=true` to remove the overhead.
```

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[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002030 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001288 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002847 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003148 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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testing was 0.001412 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
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[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003727 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002246 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004564 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001751 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001245 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001677 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002000 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002329 seconds.
You can set `force_col_wise=true` to remove the overhead.
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[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001287 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
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[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002621 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001155 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002544 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004238 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001499 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
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[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.000957 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
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[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003452 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001963 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
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testing was 0.002670 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001110 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003780 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002685 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001949 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002901 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001151 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001152 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001285 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 500, number of negative: 820
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testing was 0.003717 seconds.
You can set `force_col_wise=true` to remove the overhead.
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features: 23
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001764 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002097 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
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[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001017 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002309 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003285 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002425 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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the split requirements
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001538 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.003298 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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the split requirements
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001646 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003319 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003330 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001943 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
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[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001165 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002268 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003055 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002923 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001584 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004372 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002053 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
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[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001156 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003365 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003113 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003294 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001921 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002152 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004065 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001837 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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the split requirements
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.001320 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001181 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003636 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002157 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002471 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.002013 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003256 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001230 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001404 seconds.
You can set `force_col_wise=true` to remove the overhead.
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[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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testing was 0.004443 seconds.
You can set `force_col_wise=true` to remove the overhead.
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001774 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002991 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001281 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003073 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001127 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.008090 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004515 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001664 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.005950 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001542 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003624 seconds.
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You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001926 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
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[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003492 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001864 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002881 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002575 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003278 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.006110 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003825 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002690 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001477 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
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[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001977 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002288 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002231 seconds.
You can set `force_col_wise=true` to remove the overhead.
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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002483 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003437 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001325 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001835 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001930 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.002794 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001271 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002142 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001707 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.011009 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001004 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002906 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.004629 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
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[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003746 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003207 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.004907 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4821
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[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004374 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001903 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003157 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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the split requirements
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002006 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
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[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001949 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002661 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002258 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001430 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
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[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002142 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001863 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.006954 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
```

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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
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[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001555 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001500 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.002888 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.002770 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002164 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001842 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004117 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002845 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002633 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001916 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
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testing was 0.004116 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003425 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001892 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002102 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002635 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002191 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002897 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.002738 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002105 seconds.
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You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002232 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001060 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
```

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[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002866 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001618 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.006786 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001486 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003962 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001261 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.005117 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.002805 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001278 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004114 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002938 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001584 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002432 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
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[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001919 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002423 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003375 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001557 seconds.
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You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002569 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001644 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003609 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001660 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001760 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001173 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003937 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001081 seconds.
You can set `force_col_wise=true` to remove the overhead.
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[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001253 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002693 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004876 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002271 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002375 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001881 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.002983 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.007908 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002352 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001926 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001323 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003024 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002270 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002517 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001060 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002154 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003888 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004575 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
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[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001179 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001533 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001633 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002851 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002729 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001386 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002852 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002975 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001280 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001444 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001155 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002922 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
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the split requirements
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
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testing was 0.003602 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002686 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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the split requirements
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001500 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.005465 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003153 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003680 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
```

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[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001517 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
```

```
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001871 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
```

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[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004194 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001721 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001214 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001660 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002051 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
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[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001351 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.007614 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001195 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001759 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001109 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001349 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002922 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.000984 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.000956 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002099 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002039 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003246 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002604 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004501 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002397 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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the split requirements
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001594 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001637 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001810 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.006165 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001492 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001493 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003196 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002528 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.001511 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
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[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002391 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001298 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004213 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001617 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001563 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003657 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003827 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
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testing was 0.002159 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002460 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
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features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.003108 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001778 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002773 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
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testing was 0.002887 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001458 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001705 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.004419 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001517 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003903 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
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the split requirements
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001815 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001816 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001280 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001075 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
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[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001524 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.004783 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.005003 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
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[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003025 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001689 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.003426 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.000969 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.002126 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003854 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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the split requirements
[LightGBM] [Warning] Unknown parameter: min_child_wight
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.003202 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.005302 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 500, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001815 seconds.
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You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001778 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4818
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
the split requirements
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] Stopped training because there are no more leaves that meet
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[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
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You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4819
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
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[LightGBM] [Info] Start training from score -0.496698
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 499, number of negative: 820
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.001436 seconds.
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You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4824
[LightGBM] [Info] Number of data points in the train set: 1319, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378317 -> initscore=-0.496698
[LightGBM] [Info] Start training from score -0.496698
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Info] Total Bins 4821
[LightGBM] [Info] Number of data points in the train set: 1320, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378788 -> initscore=-0.494696
[LightGBM] [Info] Start training from score -0.494696
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[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Info] Number of positive: 624, number of negative: 1025
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of
testing was 0.000998 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 4853
[LightGBM] [Info] Number of data points in the train set: 1649, number of used
features: 23
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.378411 -> initscore=-0.496298
[LightGBM] [Info] Start training from score -0.496298
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Best hyperparameters are {'colsample_bytree': 0.7674987273808446,
'learning_rate': 0.04054204353470754, 'max_depth': 6, 'min_child_samples': 13,
'min_child_wight': 3.8085531585873613, 'n_estimators': 591, 'num_leaves': 24,
'reg_alpha': 0.808981408982322, 'reg_lambda': 1.7616020917249615, 'subsample':
0.8733779612690568}
```

```
Best score is: 0.9425523509159373
```

[82]:
```
y_pred_lgbm = model_search_lgbm.predict(X_test)
accuracy_lgbm = accuracy_score(y_test, y_pred_lgbm)
y_prods_lgbm = model_search_lgbm.predict_proba(X_test)[:,1]
```

```
[LightGBM] [Warning] Unknown parameter: min_child_wight
[LightGBM] [Warning] Unknown parameter: min_child_wight
```

[83]:
```
print(f"Accuracy {accuracy_lgbm}")
```

```
Accuracy 0.9265536723163842
```

[84]:
```
# Matriz confusion
cm_lgbm = confusion_matrix(y_test, y_pred_lgbm)
print(f"\nMatrix de confusion:")
print(cm_lgbm)
```

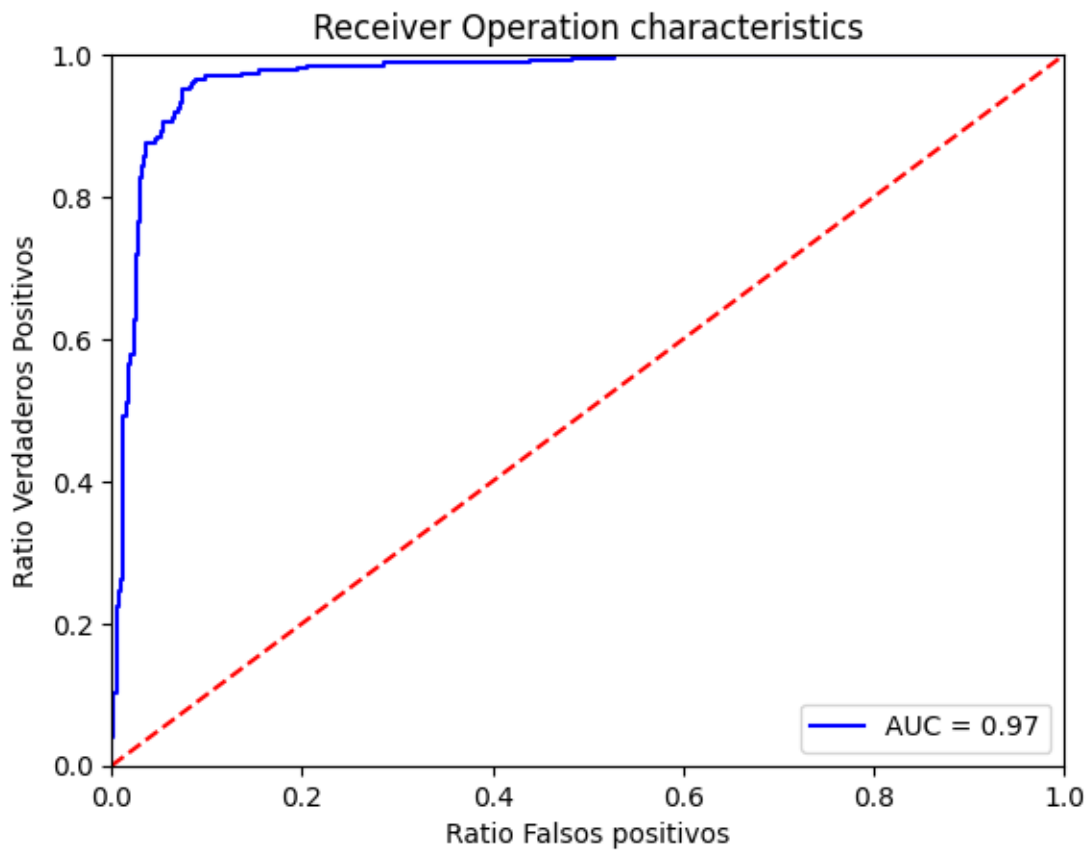
```
Matrix de confusion:
[[408  32]
 [ 20 248]]
```

[85]:
```
# Reporte de classificacion
print(f"\nReporte de clasificacion:")
print(classification_report(y_test, y_pred_lgbm))
```

```
Reporte de clasificacion:
              precision    recall  f1-score   support

           0       0.95      0.93      0.94       440
           1       0.89      0.93      0.91       268

    accuracy                           0.93       708
   macro avg       0.92      0.93      0.92       708
weighted avg       0.93      0.93      0.93       708
```

[86]:
```
## Area bajo la curva (AUC)
fpr, tpr, threshold = roc_curve(y_test, y_prods_lgbm)
roc_auc = roc_auc_score(y_test, y_prods_rf)

# plot

plt.title("Receiver Operation characteristics")
plt.plot(fpr, tpr, 'b', label="AUC = %0.2f" % roc_auc)
plt.legend(loc = "lower right")
plt.plot([0, 1], [0, 1], 'r--')
plt.xlim([0,1])
```

```
plt.ylim([0,1])
plt.ylabel("Ratio Verdaderos Positivos")
plt.xlabel("Ratio Falsos positivos")
plt.show()
```



1.5.7 Ensemble

[ ]:

[96]:
```
tree = DecisionTreeClassifier(criterion = 'entropy', max_depth = 2,␣
 ↪min_samples_split = 2)
lg = LogisticRegression(C=3.2, penalty= 'l2', random_state=23)
xgb = XGBClassifier(learning_rate = 0.01, max_depth = 3, subsample = 0.5,␣
 ↪random_state=23)
knn = KNeighborsClassifier(n_neighbors = 5)
#svm = SVC(kernel='linear', random_state=23)
rf = RandomForestClassifier(max_depth = None, max_features = 0.12,
                            min_samples_leaf = 1, n_estimators = 452,␣
 ↪min_samples_split=2, random_state=23)
```

```
my_models = [
    ('Log_Reg', lg),
    ('KNN', knn),
    ('tree', tree),
    #('SVM', svm),
    ('xgb', xgb),
    ('RF', rf)
]
```

[97]:
```
ensemble = StackingClassifier(estimators=my_models,␣
↪final_estimator=LogisticRegression(), cv=5)
```

[98]:
```
ensemble.fit(X_train, y_train)
```

[98]:
```
StackingClassifier(cv=5,
                   estimators=[('Log_Reg',
                                LogisticRegression(C=3.2, random_state=23)),
                               ('KNN', KNeighborsClassifier()),
                               ('tree',
                                DecisionTreeClassifier(criterion='entropy',
                                                       max_depth=2)),
                               ('xgb',
                                XGBClassifier(base_score=None, booster=None,
                                              callbacks=None,
                                              colsample_bylevel=None,
                                              colsample_bynode=None,
                                              colsample_bytree=None,
                                              device=None,
                                              early_stopping_round…
                                              max_cat_threshold=None,
                                              max_cat_to_onehot=None,
                                              max_delta_step=None, max_depth=3,
                                              max_leaves=None,
                                              min_child_weight=None,
                                              missing=nan,
                                              monotone_constraints=None,
                                              multi_strategy=None,
                                              n_estimators=None, n_jobs=None,
                                              num_parallel_tree=None, …)),
                               ('RF',
                                RandomForestClassifier(max_features=0.12,
                                                       n_estimators=452,
                                                       random_state=23))],
                   final_estimator=LogisticRegression())
```

[99]:
```
ensemble_pred_test = ensemble.predict(X_test)
```

[100]:
```
accuracy_ensemble = accuracy_score(y_test, ensemble_pred_test)
```

[101]:
```
print(f"Accuracy {accuracy_ensemble}")
```

```
Accuracy 0.9293785310734464
```

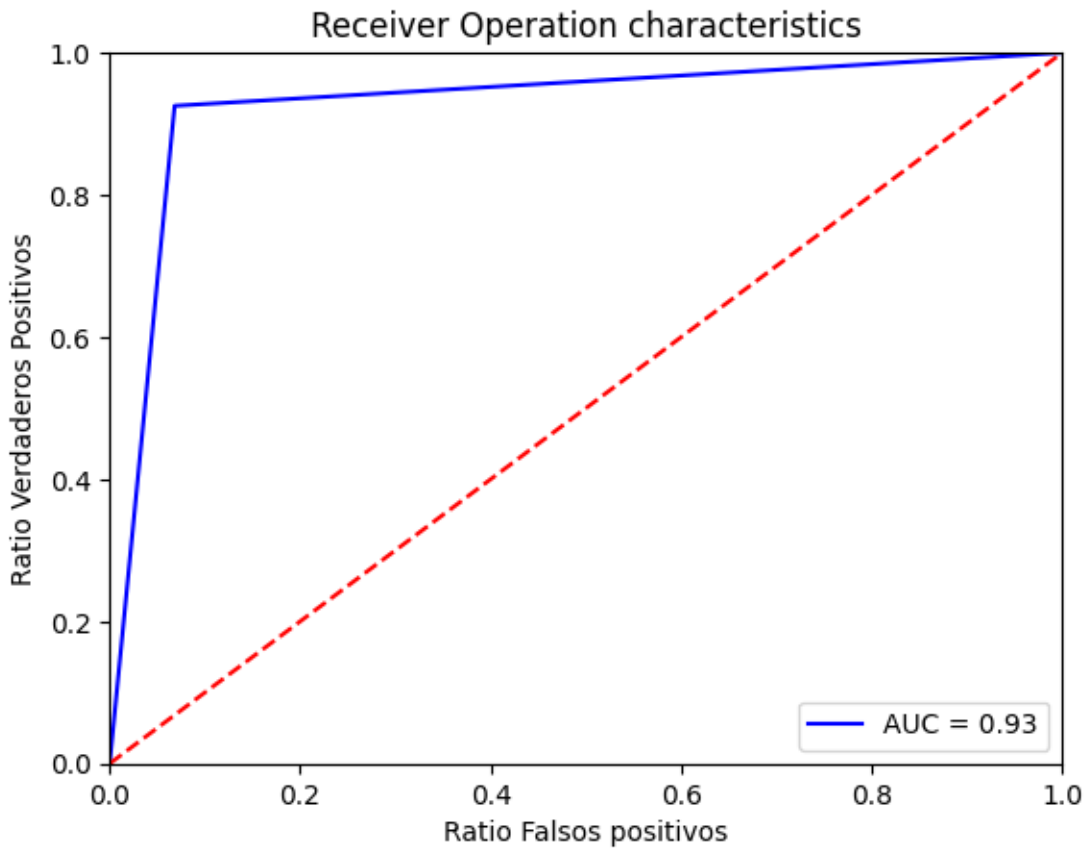
[102]:
```
# Matriz confusion
cm_ensemble = confusion_matrix(y_test, ensemble_pred_test)
print(f"\nMatrix de confusion:")
print(cm_ensemble)
```

```
Matrix de confusion:
[[410  30]
 [ 20 248]]
```

[103]:
```
# Reporte de classificacion
print(f"\nReporte de clasificacion:")
print(classification_report(y_test, ensemble_pred_test))
```

```
Reporte de clasificacion:
              precision    recall  f1-score   support

           0       0.95      0.93      0.94       440
           1       0.89      0.93      0.91       268

    accuracy                           0.93       708
   macro avg       0.92      0.93      0.93       708
weighted avg       0.93      0.93      0.93       708
```

[104]:
```
## Area bajo la curva (AUC)
fpr, tpr, threshold = roc_curve(y_test, ensemble_pred_test)
roc_auc = roc_auc_score(y_test, ensemble_pred_test)

# plot

plt.title("Receiver Operation characteristics")
plt.plot(fpr, tpr, 'b', label="AUC = %0.2f" % roc_auc)
plt.legend(loc = "lower right")
plt.plot([0, 1], [0, 1], 'r--')
plt.xlim([0,1])
plt.ylim([0,1])
plt.ylabel("Ratio Verdaderos Positivos")
plt.xlabel("Ratio Falsos positivos")
plt.show()
```

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